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MIL-STD 2045-17508-1
29 July 1994

MILITARY STANDARD

Information Technology
DOD Standardized Profiles AFT1(D)

FILE TRANSFER, ACCESS, AND MANAGEMENT

Part 1: Specification of ACSE, Presentation, and Session Protocols for the use by FTAM

AMSC N/A

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Foreword

This military standard is approved for use by all Departments and Agencies of the Department of Defense (DOD).

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this MIL-STD should be addressed to the:

Joint Interoperability and Engineering Organization (JIEO)
ATTN: TBBF
Fort Monmouth, New Jersey 07703-5613

by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this MIL-STD or by memorandum.

This MIL-STD 2045-17508 DOD Standardized Profile (DSP) is a functional standard produced by the Data Communications Protocol Standards (DCPS) Technical Management Panel (DTMP). DTMP functional standards are functional groupings of base standards. Referenced base standards may be commercial, DOD or de facto standards, although International Standards (produced by ISO, CCITT (now known as ITU-T), and other bodies) are preferred when possible.

This Defense Standardized Profile (DSP) is a functional DOD Data Communications Protocol Standard (DCPS) produced by the DCPS Technical Management Panel (DTMP). The MIL-STD-2045 document series was established within the DCPS Standardization Area to allow for the enhancement of commercial standards or the development of standards that are unique to DoD.

The MIL-STD-2045-10000 series, MIL-STD-2045-10000 to MIL-STD-2045-19999 inclusive, will be used to describe how DOD will implement commercial, international, national, federal, or military standards within the functional profile concept to provide required network services. The Government Open Systems Interconnection Profiles (GOSIP) will serve as the base for developing the 10000 series with DOD enhancements, unique military standards, and interim standards being used only when necessary.

The MIL-STD-2045-20000 series, MIL-STD-2045-20000 to MIL-STD-2045-29999 inclusive, will be used to describe DOD enhancements and extensions to existing commercial, international, national, or federal standards.

The MIL-STD-2045-30000 series, MIL-STD-2045-30000 to MIL-STD-2045-39999 inclusive, will be used to describe protocols and services unique to DOD that will not be supported by commercial, international, national, or federal standards.

The MIL-STD-2045-40000 series, MIL-STD-2045-40000 to MIL-STD-2045-49999 inclusive, will be used to document interim standards. Interim standards document protocols and services needed by DOD until these protocols and services are described in either a GOSIP or a MIL-STD-2045-20000 or -30000 series standard.

The MIL-STD-2045-50000 series, MIL-STD-2045-50000 to MIL-STD-2045-59999 inclusive, will be used to describe how DOD will implement commercial, international, national, federal, or military standards within the functional profile concept to provide required network services. The Government Open Systems Interconnection Profiles (GOSIP) will serve as the base for developing the 50000 series with DOD enhancements, unique military standards, and interim standards being used only when necessary. The difference between MIL-STD-2045-10000 series and the MIL-STD-2045-50000 series is that the 50000 series are interim profiles.

Specific details and instructions for establishing a MIL-STD-2045 document, as well as profile development guidelines, are documented in MIL-HDBK-829. DTMP Working Groups shall be responsible for DSP development and informal Service or Agency coordination; the DTMP Plenary shall be responsible for final review and approval.

This document was produced as an outgrowth of a requirement established for transmitting digital imagery and imagery related products using the National Imagery Transmission Format Standard (NITFS), and is intended to be a file transfer profile for end systems to communicate over DOD or commercial circuits.

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This part of MIL-STD 2045-17508 contains two normative annexes and one informative annex.

Annex A (normative)	DSPICS Requirements List for ACSE, Presentation and Session
Annex B (normative)	Amendments and corrigenda
Annex C (informative)	Concluding Material

This document is a DSP suitable for transmission of digital imagery using a file transfer protocol. It is in addition to the current Taxonomy and Framework for International Standardized Profiles.

The current technical content of the document has been derived from ISO 8571, parts 1 through 4. However, this document is based on DOD requirements. Differences between the content of this document and the base standards may exist.

For DOD acquisition purposes, where such differences exist, this DSP shall be the controlling document.

The Preparing Activity for this standard is the Data Communication Protocol Standards Technical Management Panel (DTMP). The custodians for the document are identified in the Defense Standardization Program, "Standardization Directory (SD-1)" and are classified in the Federal Supply Classification (FSC) system under Data Communication Protocol Standards (DCPS). Additional information can be obtained from:

Joint Interoperability and Engineering Organization
ATTN: DTMP Chairman
Ft. Monmouth, New Jersey 07703-5613

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Introduction

This DOD Standardized Profile (DSP) is defined within the context of functional standardization, in accordance with the principles specified by ISO/IEC TR 10000, "Framework and Taxonomy of International Standardized Profiles" and MIL-HDBK-829. The context of functional standardization is one part of the overall field of Information Technology (IT) standardization activities - covering base standards, profiles, and registration mechanisms. A profile defines a combination of base standards that collectively perform a specific well-defined IT function. Profiles standardize the use of options and other variations in the base standards to promote system interoperability and to provide a basis for the development of uniform, internationally recognized system tests.

One of the most important roles for a DSP is to serve as the basis for the development of recognized tests. DSPs also guide implementors in developing systems that fit the needs of the US Department of Defense (DOD). DSPs are produced not simply to 'legitimize' a particular choice of base standards and options, but to promote real system interoperability. The development and widespread acceptance of tests based on this and other DSPs is crucial to the successful realization of this goal.

The base standards of this DSP include Open Systems Interconnection (OSI) Layer Standards from the OSI Reference Model.

The specifications in this part of MIL-STD 2045-17508 specify how the Association Control Service Element, the Presentation Layer and the Session Layer standards shall be used to provide FTAM functions.

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Information Technology - DOD Standardized Profile (DSP) AFT1(D) - File Transfer, Access, and Management - Part 1: Specification of ACSE, Presentation, and Session Protocols for use by FTAM

1 Scope

1.1 General

This part of MIL-STD 2045-17508 specifies how the Association Control Service Element, the Presentation Layer, and the Session Layer standards shall be used to provide the FTAM functions for the Defense Standardized Profiles MIL-STD 2045-17508 (see also figure 1). These specifications are therefore the common basis for the application functions as defined in MIL-STD 2045-17508-2, MIL-STD 2045-17508-3, MIL-STD 2045-17508-6, and any of the subsequent parts that might be developed.

1.2 Position within the taxonomy

This part of MIL-STD 2045-17508 is the first part of a multi-part DSP for AFT1n(D) File Transfer, Access and Management. The multi-part DSP consists of the following parts:

Part 1: Specification of ACSE, Presentation, and Session Protocols for the use by FTAM

Part 2: Definition of document types, constraint sets and syntaxes

Part 3: Simple File Transfer Service (unstructured)

Part 6: File Management Service

It may be combined with any T-Profiles (see ISO/IEC TR 10000) specifying the OSI connection-mode transport service.

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1.3 Scenario

The model used is one of two end systems running an end-to-end association using the ACSE, Presentation and Session services and protocols (see figure 1).

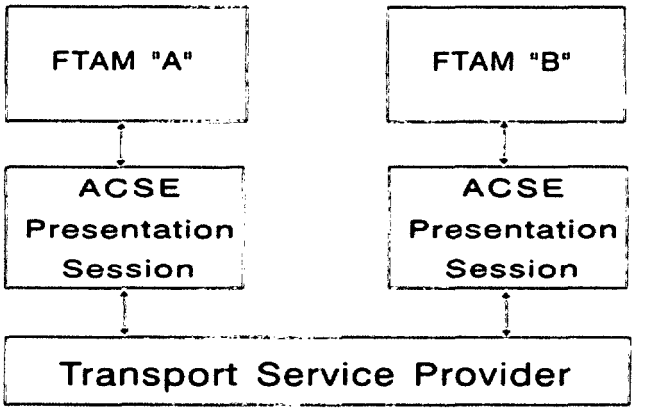


Figure 1 - Model of the supportive layers

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of this part of MIL-STD 2045-17508. At the time of publication, the editions indicated were valid. All documents are subject to revision, and parties to agreements based on this part of MIL-STD 2045-17508 are warned against automatically applying any more recent editions of the documents listed below, since the nature of references made by DSPs to such documents is that they may be specific to a particular edition.

Amendments and corrigenda to the base standards referenced: See annex B for a complete list of these documents which are used in this part of MIL-STD 2045-17508.

2.1 Government Documents:

2.1.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

MIL-HDBK 829: July 1994 *Guidelines for Developing Data Communications Profiles.*

DOD activities may obtain copies of DOD directives through their own publication channels or from the DOD Single

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Stock Point, Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094. Other federal agencies and the public may purchase copies from the U.S. Department of Commerce, National Technical Information Service, 5285 Port Royal Road, Springfield, VA 22161.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

None.

Copies of Federal Information Processing Standards (FIPS) are available to Department of Defense activities from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120-5099. Others must request copies of FIPS from the National Technical Information Services, 5285 Port Royal, Springfield, VA 22161-2171.

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation.

2.2.1 Profiles.

ISO/IEC ISP 10607-2: 1990, Information technology - International Standardized Profiles AFTnn - File Transfer, Access and Management - Part 2:Definition of document types, constraint sets and syntaxes.

ISO/IEC ISP 10607-3: 1990, Information technology - International Standardized Profiles AFTnn - File Transfer, Access and Management - Part 3:AFT11 - Simple File Transfer Service (unstructured).

DIS/IEC ISP 10607-6: 1990, Information technology - International Standardized Profiles AFTnn - File Transfer, Access and Management - Part 6:AFT3 - File Management Service.

Application for copies of these documents should be addressed to the American National Standards Institute, 11 West 42nd Street, NY, NY 10036 or to ISO, Van Demonstrate 94, 1013 CN Amsterdam, Netherlands.)

2.2.2 Base Standards:

ISO 8571-1: 1988, Information processing systems - Open Systems Interconnection - File Transfer, Access and Management - Part 1: General introduction.

ISO 8571-2: 1988, Information processing systems - Open Systems Interconnection - File Transfer, Access and Management - Part 2: Virtual Filestore definition.

ISO 8571-3: 1988, Information processing systems - Open Systems Interconnection - File Transfer, Access and Management - Part 3: File Service Definition.

ISO 8571-4: 1988, Information processing systems - Open Systems Interconnection - File Transfer, Access and Management - Part 4: File Protocol Specification.

ISO 8571-5: 1990, Information processing systems - Open Systems Interconnection - File Transfer, Access and Management - Part 5: Protocol Implementation Conformance Statement Proforma.

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2.2.3 Other Non-Government documents, drawings, and publications.

The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation:

ISO 8327: 1987, Information processing systems - Open Systems Interconnection - Basic Connection Oriented Session Protocol Specification.

ISO 8327/Add.2—[1], Information processing systems - Open Systems Interconnection - Basic Connection Oriented Session Protocol Specification - ADDENDUM 2: Incorporation of unlimited user data.

ISO 8650: 1988, Information processing systems - Open Systems Interconnection - Protocol specification for the Association Control Service Element.

ISO 8823: 1988, Information processing systems - Open Systems Interconnection - Connection Oriented Presentation Protocol Specification.

ISO 8824: 1987, Information processing systems - Open Systems Interconnection - Specification of Abstract Syntax Notation One (ASN.1).

ISO 8825: 1987, Information processing systems - Open Systems Interconnection - Specification of Basic Encoding Rules for Abstract Syntax Notation One (ASN.1).

(Application for copies of these documents should be addressed to the American National Standards Institute, 11 West 42nd Street, NY, NY 10036 or to ISO, Van Demonstrate 94, 1013 CN Amsterdam, Netherlands.)

[1]To be published

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3 Definitions

For the purposes of this part of MIL-STD 2045-17508, the following definitions apply.

Terms used in this part of MIL-STD 2045-17508 are defined in the referenced base standards.

In addition, the following terms are defined.

3.1 General

Interwork: to be able to communicate to satisfy the intent of the Initiator.

4 Abbreviations and Acronyms

ACSE	Association Control Service Element
AE	Application Entity

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AFT	Profile sub-class: File Transfer, Access and Management
AP	Application Process
APDU	Application Protocol Data Unit
ARP	Abnormal Release Provider
ARU	Abnormal Release User
ASN.1	Abstract Syntax Notation One
CP	Connect Presentation
CPC	Connect Presentation Context
CPR	Connect Presentation Reject
DSP	Defense Standardized Profile
FADU	File Protocol Data Unit
FTAM	File Transfer, Access and Management
ISP	International Standardized Profile
OSI	Open Systems Interconnection
PCI	Protocol Control Information
PDU	Protocol Data Unit
PDV	Presentation Data Value
PICS	Protocol Implementation Conformance Statement
PPDU	Presentation Protocol Data Unit
RS	Resynchronize
SPDU	Session Protocol Data Unit

Support level for protocol features:

m	supported
o	optionally supported
c	conditionally supported
x	excluded
i	outside the scope
-	not applicable

5 Conformance

This part of MIL-STD 2045-17508 states requirements upon implementations to achieve interworking. A claim of conformance to this part of MIL-STD 2045-17508 is a claim that all requirements in the relevant base standards are satisfied, and that all requirements in the following clauses and in annex A of this part of MIL-STD 2045-17508 are satisfied. Annex A states that relationship between these requirements and those of the base standards.

5.1 Conformance statement

For each implementation claiming conformance to this part of MIL-STD 2045-17508 an appropriate set of PICSs shall be made available stating support or non-support of each option identified in this part of MIL-STD 2045-17508.

5.2 Relationship with base standards

5.2.1 ACSE conformance

No additional requirements.

5.2.2 Presentation layer conformance

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No additional requirements.

5.2.3 Transfer syntax conformance

No additional requirements.

5.2.4 Session layer conformance

No additional requirements.

6 Association Control Service Element (ACSE)

No additional requirements.

6.1 Application entity title

No additional requirements.

6.2 Application context name

"ISO FTAM" { iso(1) 8571 application-context iso-ftam(1) } - implies the use of the ACSE and the FTAM ASE.

6.3 Abort APDU

No additional requirements.

7 Presentation layer

The support of functions and parameters for the Presentation protocol is as specified in annex A, clause A.2. An implementation's PICS may contain limitations on length or value aspects of a protocol, but is shall not contain limits more severe than those specified in this clause. An implementation conformant to this part of MIL-STD 2045-17508 may abort a connection, if the requirements as specified in 7.1 to 7.8 are violated.

7.1 Presentation context identifier

At least 3 Presentation Contexts must be supported. No additional requirements.

7.2 Transfer syntax name in PDV list

No additional requirements.

7.3 Encoding of presentation data value

No additional requirements.

7.4 User data parameter

No additional requirements.

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7.5 Encoding of CPC-type value

No additional requirements.

7.6 Presentation context definition result list

No additional requirements.

7.7 Resynchronize PPDU

No additional requirements.

7.8 Invalid encoding

No additional requirements.

8 Transfer syntax

Subclauses 8.1 through 8.5 specify rules which limit the encoding of data defined in this part of MIL-STD 2045-17508 according to the Basic Encoding Rules for Abstract Syntax Notation One. A decoder shall accept encodings that are within these limits. Other behaviors of the decoder when these limits are exceeded, are outside the scope of this part of MIL-STD 2045-17508.

8.1 ASN.1 length fields

No additional requirements.

8.2 Integer type values

No additional requirements.

8.3 Bitstring type values

No additional requirements.

8.4 External Data Type

No additional requirements.

8.5 Constructed encoding

No additional requirements.

9 Session Layer

The support of functions and parameters for the Session protocol is as specified in annex A, clause A.3. An implementation's PICS may contain limitations on length or value aspects of a protocol, but it shall not contain limits more severe than those specified in this clause. An implementation conformant to this part of MIL-STD 2045-17508 may abort a connection, if the requirements as specified in 9.1 to 9.8 are violated.

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9.1 Session version

No additional requirements.

9.2 Extended concatenation

No additional requirements.

9.3 Session segmenting

No additional requirements.

9.4 Minor synchronize

The Session minor synchronize functional unit shall be implemented.

9.5 Resynchronize

The Session resynchronize functional unit shall be implemented.

9.6 Re-use of transport connection

No additional requirements.

9.7 Use of transport expedited

No additional requirements.

9.8 Receipt of invalid SPDUs

No additional requirements.

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ANNEX A (normative)

DSPICS REQUIREMENTS LIST (DPRL)

for ACSE, Presentation and Session

A.1 Introduction

This document provides the DoD Standardized Profile Implementation Conformance Statements (DSPICS) Requirements List (DPRL) for implementations of the MIL-STD 2045-17508. The DSPICS for an implementation is generated by completing the DPRL in accordance with the following instructions.

An implementation shall satisfy the mandatory conformance requirements of the base standards referenced in this profile.

An implementation's completed DPRL is called the DSPICS. The DSPICS states which capabilities and options of the protocol have been implemented. The following can use the DSPICS:

(a) the protocol implementor, as a checklist to reduce the risk of failure to conform to the standard through oversight.

(b) the supplier and acquirer or potential acquirer of the implementation, as a detailed indication of the capabilities of the implementation, stated relative to the common basis for understanding provided by the standard DSPICS proforma.

(c) the user or potential user of the implementation, as a basis for initially checking the possibility of inter-working with another implementation (note that, while inter-working can never be guaranteed, failure to internetwork can often be predicted from incompatible DSPICSs).

(d) a protocol tester, as the basis for selecting appropriate tests against which to assess the claim for conformance of the implementation.

A.1.1 Notation

The following notations and symbols from ISP 10607 are used in the DPRL to indicate the basis for selection of options:

D - As defined in the base standard
P - As defined in MIL-STD 17508
_ - underline indicates change from ISP 10607 selection

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Example:

Protocol Element/Parameter	D	P	Range of Values or Reference
FTAM unstructured text abstract syntax	o	o	

The following notations and symbols from MIL-HDBK 829, which references ISO/IEC TR 10000-1 and -2, are used in the DPRL to indicate the status of features:

Status Symbols

m	- mandatory
m.<n>	- support of every item of the group labeled by the same numeral <n> required, but only one is active at a time
o	- optional
o.<n>	- optional, but support of at least one of the group of options labeled by the same numeral <n> is required
c	- conditional
-	- non-applicable (i.e. logically impossible in the scope of the profile)
x	- excluded or prohibited
i	- out of scope of profile (left as an implementation choice)

In addition, the symbol "•" is used to indicate an option whose status is not constrained by the profile (status in the base standard). The o.<n> notation is used to show a set of selectable options (i.e., one or more of the set must be implemented) with the same identifier <n>.

Two character combinations may be used for dynamic conformance requirements. In this case, the first character refers to the static (implementation) status, and the second refers to the dynamic (use); thus "mo" means "mandatory to be implemented, optional to be used."

A.1.2 Support level: To specify the support level of protocol features for this part of MIL-STD 2045-17508, the following terminology is defined and is applicable to this DSP.

supported; m: Any feature denoted by "m" is mandatory or optional in the base standard. That feature shall be supported, i.e. its syntax and procedures shall be implemented as specified in the base standard or in this part of MIL-STD 2045-17508 by all implementations claiming conformance to this part MIL-STD 2045-17508.

However, it is not a requirement that this feature shall be used in all instances of communication, unless mandated by the base standard or stated otherwise in this part of MIL-STD 2045-17508.

For fully supported attributes, this implies that at least the minimum range of attribute values, as defined in ISO 8571-2, shall be supported unless stated otherwise in this part of MIL-STD 2045-17508.

optionally supported; o: Any feature denoted by "o" is left to the implementation as to whether that feature is implemented or not.

If an attribute group with a support level of "o" is chosen to be implemented, then all the attributes in this group that are classified as "m" shall be supported.

If a parameter is optionally supported, then the syntax shall be implemented, but it is left to each implementation whether the procedures are implemented or not.

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When receiving an optional parameter which is not subject of negotiation and is not supported by the Receiver, the Receiver shall at least inform the Sender by information diagnostic and interworking shall not be disrupted.

conditionally supported; c: Any feature denoted by "c" shall be supported under the conditions specified in this part of MIL-STD 2045-17508. If these conditions are not met, the feature is outside the scope of this part of MIL-STD 2045-17508.

excluded; x: Any feature denoted by "x" is excluded in this part of MIL-STD 2045-17508, i.e. it shall not be implemented.

outside the scope; i: Any feature denoted by "i" is outside the scope of this part of MIL-STD 2045-17508, i.e. it may be ignored, and will therefore not be subject of an MIL-STD conformance test. However, the syntax of all parameters of supported PDUs shall be implemented, even if the procedures are not (i.e. the Receiver shall be able to decode the PDU).

not applicable; - : Any feature denoted by "-" is not defined in the context where it is mentioned, e.g. a parameter which is not part of the respective PDU. The occurrence of n"not applicable" features is mainly due to the format of the tables in the DSPICS Requirements List.

Notes:

1 For features which are optional in the base standard, conformant implementations shall be able to interwork with other implementations not supporting this feature.

2 The support of a feature can be conditional, depending on the support of a class of features to which it belongs, e.g. a parameter in a PDU, a PDU in a functional unit.

Notations for Conditional Status

The following predicate notations are used:

<predicate>:: This notation introduces a group of items, all of which are conditional on <predicate>.

<predicate>: This notation introduces a single item which is conditional on <predicate>.

In each case, the predicate may identify a profile feature, or a boolean combination of predicates. ("^" is the symbol for logical negation.)

<index>: This predicate symbol means that the status following it applies only when the DSPICS states that the features identified by the index are supported. In the simplest case, <index> is the identifying tag of a single DSPICS items. The symbol <index> also may be a Boolean expression composed of several indices.

<index>:: When this group predicate is true, the associated clause should be completed.

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Notations used in the Protocol Feature Column

<r> Symbol used to denote the receiving system.
<t> Symbol used to denote the transmitting system.

Support Column Symbols

The support of every item as claimed by the implementor is stated by circling the appropriate answer (Yes, No, or N/A) in the support column:

Yes Supported by the implementation.
No Not supported by the implementation.
N/A Not applicable.

Base standard requirements are shown using the equivalent notations in upper case (e.g., M, O, X).

A.1.3 Footnotes

Footnotes to the proforma are indicated by superscript numerals. The footnote appears on the page of the first occurrence of the numeral. Subsequent occurrences of a numeral refer to the footnote of the first occurrence.

A.1.4 Instructions for Completing the DPRL

A DSP implementor shows the extent of compliance to a DSP by completing the DPRL; that is, compliance to all mandatory requirements and the options that are not supported are shown. The resulting completed DPRL is called a DSPICS. Where this profile refines the features of the base standards, the requirements expressed in this DPRL shall be applied (as indicated in DPRL items with no "Profile Support" column) to constrain the allowable responses in the base standard DSPICS proforma. When this profile makes additional requirements, the "Profile Support" column for such DPRLs shall be completed. In this column, each response shall be selected either from the indicated set of responses, or it shall comprise one or more parameter values as requested. If a conditional requirement is inapplicable, use the Not Applicable (NA) choice. If a mandatory requirement is not satisfied, exception information must be supplied by entering a reference Xi, where i is a unique identifier, to an accompanying rationale for the noncompliance. When the profile requirement is expressed as a two-character combination (as defined in A.1.1 above), the response shall address each element of the requirement; e.g., for the requirement "mo," the possible compliant responses are "yy" or "yn."

A.2 Standards Referenced

This AF profile specifies how the Association Control Service Element, the Presentation Layer and the Session Layer standards shall be used to provide FTAM functions. It uses the following standards:

ISO 8650 Protocol specification for the Association Control Service Element.
ISO 8327 Basic Connection Oriented Session Protocol Specification.
ISO 8823 Connection Oriented Presentation Protocol Specification.

A.3 DSPICS Requirements List

The following tables give the DPRL for specification of ACSE, Presentation, and Session protocols for the use by FTAM.

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A.3.1 General Information

A.3.1.1 Implementation Identification

Supplier	
Contact point for queries about the profile	
Implementation Name(s) and Version(s)	
Date of statement	
Other Information: Machine Name, Operating System, System Name	

A.4 Association Control Service Element

Table A.4 - Association Control Service Element protocol

No additional requirements.

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A.5 Presentation layer

Table A.5 - Presentation Protocol

Protocol Element/Parameter	D	P	Range of Values or Reference
Defect Solutions and Amendments These defect reports to ISO 8823 have been resolved. They are, therefore, considered as part of ISO 8823 on which this part of MIL-STD 2045-17508 is based.			See annex B
FUNCTIONAL UNITS:			
kernel	m	m	
context management	o	i	
context restoration	o	i	
			OBJECT IDENTIFIER values:
ABSTRACT SYNTAXES: FTAM PCI	m	m	{ iso standard 8571 abstract-syntax(2) ftam-pci(1)}
FTAM FADU	o	c	{iso standard 8571 abstract-syntax(2) ftam-fadu(2)}
FTAM unstructured text abstract syntax	o	o	{iso standard 8571 abstract-syntax(2) unstructured-text(3)}
FTAM unstructured binary abstract syntax	o	m	{iso standard 8571 abstract-syntax(2) unstructured-binary(4)
NBS abstract syntax AS1	-	c	{iso identified-organization oiw(14) ftamsig(5) abstract-syntax(2) nbs-as1(1)}
NBS file-directory entry abstract syntax (NBS-AS2)	-	c	{iso identified-organization oiw(14) ftamsig(5) abstract-syntax(2) nbs-as2(2)}
NBS random access node name abstract syntax (NBS-AS3)	-	c	{iso identified-organization oiw(14) ftamsig(5) abstract-syntax(2) nbs-node-name(3)}
NBS random binary access file abstract syntax (NBS - AS4)	-	c	{iso identified-organization oiw(14) ftamsig(5) abstract-syntax(2) nbs-random-binary(4)}
NBS simple text abstract syntax (NBS - AS5)	-	c	{iso identified-organization oiw(14) ftamsig(5) abstract-syntax(2) nbs-simple-text(5)}
INTAP abstract syntax AS1	-	c	{iso member-body 392 ftam(10) abstract-syntax(3) intap-as1(1)}
(abstract syntax for ACSE)	m	m	joint-iso-ccitt association-control(2) abstract-syntax(1) apdus(0) version1(1)}

NOTE - The support requirements for the conditional abstract syntaxes depend on the constraint sets and document types which are supported by FTAM (see annex A, clause A.9 of MIL-STD 2045-17508-3).

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A.6 Session layer

Table A.6 - Session protocol

Protocol Element/Parameter	D	P	Range of Values or Reference
Defect Solutions and Amendments These defect reports to ISO 8327 have been resolved. They are, therefore, considered as part of ISO 8327/Add.2 on which this part of MIL-STD 2045-17508 is based.			see annex B
FUNCTIONAL UNITS:			
kernel	m	m	
half - duplex	o	i	
duplex	o	m	
expedited data	o	i	
minor synchronize	o	m	see 9.4 in ISP 10607-1
major synchronize	o	m	
resynchronize	o	m	see 9.5 in ISP 10607-1
activity management	o	i	
negotiated release	o	i	
capability data exchange	o	i	
exceptions	o	i	
typed data	o	i	
TOKENS:			
release	o	i	
major/activity	o	i	
synchronize-minor	o	m	see 9.4 in ISP 10607-1
data	o	i	
SESSION FEATURES:			
basic concatenation	m	m	
extended concatenation	o	o	see 9.2 in ISP 10607-1
segmenting	o	o	see 9.3 in ISP 10607-1
re-use of transport connection	o	m	see 9.6 in ISP 10607-1
use of transport expedited	o	o	see 9.7 in ISP 10607-1
SPDUs AND PARAMETERS:			see 9.8 in ISP 10607-1
Connection Establishment:			
CONNECT	m	m	
connection identifier	o	i	
connect/accept item	o	m	
protocol options	m	m	
TSDU maximum size	o	o	
version number	m	m	see 9.1 in ISP 10607-1
initial serial number	o	o	
token setting item	o	o	

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Table A.6 (continued)

Protocol Element/Parameter	D	P	Range of Values or Reference
session user requirement			
calling session selector	o	m	
called session selector	o	mo	
data overflow	o	mo	
user data	o	i	
extended user data	o	m	
	o	m	
ACCEPT			
enclosure item	m	m	
connection identifier	o	i	
connect/accept item	o	i	
protocol options	o	m	
TSDU maximum size	m	m	
version number	o	o	
initial serial number	m	m	see 9.1 in ISP 10607-1
token setting item	o	o	
token item	o	o	
session user requirements	o	i	
calling session selector	o	m	
responding session selector	o	mo	
user data	o	mo	
	o	m	
REFUSE			
enclosure item	m	m	
connection identifier	o	i	
session user requirements	o	i	
transport disconnect	o	m	
version number	o	m	see 9.6 in ISP 10607-1
reason code	m	m	see 9.1 in ISP 10607-1
	o	m	
OVERFLOW ACCEPT			
CONNECT DATA OVERFLOW	o	i	
	o	i	
Connection Release:			
FINISH			
enclosure item	m	m	
transport disconnect	o	i	
user data	o	m	see 9.6 in ISP 10607-1
	o	m	
DISCONNECT			
enclosure item	m	m	
user data	o	i	
	o	m	
NOT FINISHED			
	o	i	
ABORT			
enclosure item	m	m	
transport disconnect	o	i	
reflect parameter values	m	m	
user data	o	o	
	o	m	

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Table A.6 (continued)

Protocol Element/Parameter	D	P	Range of Values or Reference
ABORT ACCEPT	o	i	
Data Transfer:			
DATA TRANSFER	m	m	
enclosure item	o	o	
user information field	m	m	
EXPEDITED DATA	o	i	
TYPED DATA	o	i	
CAPABILITY DATA	o	i	
CAPABILITY DATA ACK	o	i	
Token Management:			
GIVE TOKENS	m	m	
enclosure item	o	i	
token item	o	o	
user data	o	i	
PLEASE TOKENS	o	i	
GIVE TOKENS CONFIRM	o	i	
GIVE TOKENS ACK	o	i	
Synchronize:			
MINOR SYNC POINT	o	m	see 9.4 in ISP 10607-1
enclosure item	o	i	
sync type item	o	m	
serial number	m	m	
user data	o	i	
MINOR SYNC ACK	o	m	see 9.4 in ISP 10607-1
enclosure item	o	i	
serial number	m	m	
user data	o	i	
MAJOR SYNC POINT	o	i	
MAJOR SYNC ACK	o	i	
Resynchronization:			
RESYNCHRONIZE	o	m	see 9.5 in ISP 10607-1
enclosure item	o	i	

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Table A.6 (continued)

Protocol Element/Parameter	D	P	Range of Values or Reference
token setting item	o	m	
resync type	m	m	
serial number	m	m	
user data	o	m	
RESYNCHRONIZE ACK	o	m	see 9.5 in ISP 10607-1
enclosure item	o	i	
token setting item	o	m	
serial number	m	m	
user data	o	m	
PREPARE	o	o	see 9.7 in ISP 10607-1
Exception reporting:			
EXCEPTION REPORT	o	i	
EXCEPTION DATA	o	i	
Activity Management:			
ACTIVITY START	o	i	
ACTIVITY RESUME	o	i	
ACTIVITY INTERRUPT	o	i	
ACTIVITY INTERRUPT ACK	o	i	
ACTIVITY DISCARD	o	i	
ACTIVITY DISCARD ACK	o	i	
ACTIVITY END	o	i	
ACTIVITY END ACK	o	i	

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ANNEX B (normative)

AMENDMENTS AND CORRIGENDA

International Standards are subject to constant review and revision by the ISO/IEC Technical Committee concerned. The following amendments and corrigenda are approved by ISO/IEC JTC1, but at the date of publication of this part of MIL-STD 2045-17508 they were not yet incorporated in the text of the corresponding base standards ar referenced in clause 2. The amendments and corrigenda as listed below are considered as normative references in this part of MIL-STD 2045-17508.

B.1 ACSE

ISO 8650/Cor.1:1990

B.2 Session

Solution to ISO 8327 Defects:

8326/002:ISO/IEC JTC1 SC21 N4659

8326/005:ISO/IEC JTC1 SC21 N 4660

83265/025:ISO/IEC JTC1 SC21 N 4637

8326/026:ISO/IEC JTC1 SC21 N 4638

8327/037:ISO/IEC JTC1 SC21 N 4661

8327/043:ISO/IEC JTC1 SC21 N 4663

8327/045:ISO/IEC JTC1 SC21 N 4664

8327/047:ISO/IEC JTC1 SC21 N 4665

8327/048:ISO/IEC JTC1 SC21 N 4666

B.3 FTAM

ISO 8571-1/Cor.1:1991

ISO 8571-2/Cor.1:1991

ISO 8571-3/Cor.1:1991

ISO 8571-3/Cor.2:1992

ISO 8571-4/Cor.1:1992

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ANNEX C (Informative)

CONCLUDING MATERIAL

C.1 Deviations from Base Standards/Referenced Profiles

LAYER: APPLICATION
PART 1, Clause 6.2

Protocol Element/Parameter	ISP
Application context name	"ISO 8571-4 defines a value for a simple transfer mechanism. Other values are not defined in this part of ISO/IEC 10607."

Rationale: To align MIL-STD with the NIST OIW Stable Implementors Agreements.

LAYER: PRESENTATION
PART 1, Clause 7.1

Protocol Element/Parameter	ISP
Presentation context identifier	"A conformant implementation shall encode presentation context identifiers in the range 0 to 32,767.

Rationale: To align MIL-STD with the NIST OIW Stable Implementors Agreements.

LAYER: SESSION
PART 1, Clause 9.4

Protocol Element/Parameter	ISP
Minor synchronize	"The Session minor synchronize functional unit shall be implemented if FTAM recovery or restart data transfer functional units are supported or if session resynchronize functional unit is implemented.

Rationale: To support implementation of the Restart and Recovery functional units.

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LAYER: SESSION
PART 1, Clause 9.5

Protocol Element/Parameter	ISP
Resynchronize	"The session resynchronize functional unit shall be implemented if FTAM restart data transfer functional unit is implemented.

Rationale: To support implementation of the Restart and Recovery functional units.

LAYER: PRESENTATION

Protocol Element/Parameter	Base(D)	ISP	DSP(I/R)
FTAM unstructured text abstract syntax	o	m	o

Rationale: The characteristics of the NITF digital image transferred is generally a large data file with binary-type content. This parameter was made optional in case the users decided to optionally send unstructured text.

Protocol Element/Parameter	Base(D)	ISP	DSP(I/R)
Functional Units			
minor synchronize	o	i	m
major synchronize	o	i	m
resynchronize	o	c	m

Rationale: Required to support implementation of the Restart and Recovery functional units.

Protocol Element/Parameter	Base(D)	ISP	DSP(I/R)
Tokens			
synchronize-minor	o	m	m

Rationale: Required to support the Restart and Recovery functional units.

Protocol Element/Parameter	Base(D)	ISP	DSP(I/R)
Session Features			
re-use of transport connection	o	o	m

Rationale: Required to support the Restart and Recovery functional units.

Protocol Element/Parameter	Base(D)	ISP	DSP(I/R)
Connect			
calling session selector	o	om	mo
Accept			
calling session selector	o	om	mo

Rationale: Typographical error in the ISO 10607-1.

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Protocol Element/Parameter	Base(D)	ISP	DSP(I/R)
REFUSE			
transport disconnect	o	c	m
FINISH			
transport disconnect	o	c	m
Minor Sync Point	o	c	m
Minor Sync ACK	o	c	m
Resynchronize	o	c	m
Resynchronize ACK	o	c	m

Rationale: Required to support the Restart and Recovery functional units.

C.2 Subject Term (Keyword) Listing

Communication Protocol Standards
Data Communications
Functional Profiles
Interoperability
Application
File transfer
Standards

C.3 Preparing Activity

DISA-JIEO (Project DTMP-0008-4)

C.4 Reviewing Activities

Army : SC, PT
Air Force: 13, 17, 29, 33, 90
DLA: DH
DMA: MP
DIA: DI
DOT: OST
NSA: NS
OASD: IQ, DO, IR
ODISC4: AC
NAVY: EC, CH, ND, TD, OM
USMC: MC, CG

C.5 Custodians

DISA: DC
Army: SC
Air Force: 90
Navy: OM
DIA: DI
NSA: NS

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USMC: MC
DLA: DH
Other: Joint Staff/Architecture & Integration
USSPACECOM

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-STD 2045-17508-1	2. DOCUMENT DATE *(YYMMDD)* 940729

3. DOCUMENT TITLE **FTAM Part 1: Specification of ACSE, Presentation, and Session Protocols for use by FTAM**

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

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c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) DSN *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

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